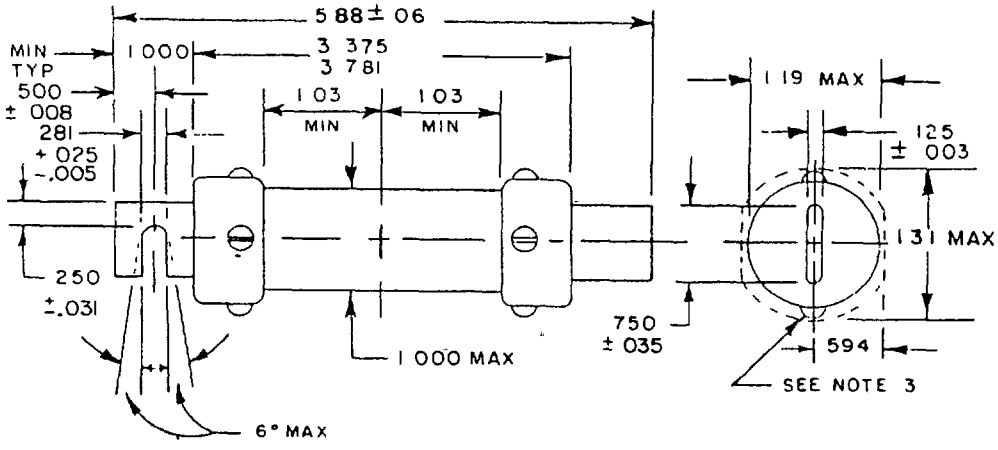
FEDERAL SPECIFICATION SHEET

FUSES, CARTRIDGE, HIGH-INTERRUPTING CAPACITY, CLASSES RK1 AND RK5 (CURRENT-LIMITING) 250 VOLTS, 61-100 AMPERES

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the fuses described herein shall consist of this document and the latest issue in effect of specification W-F-1814/GEN.



Inches	mm
.003	0.09
.005	0.13
.009	0.20
.025	0.64
.031	0.79
.035	0.89
.06	1.5
.125	3.18
.250	6.35
.281	7.14
.500	12.70
.594	15.09
.750	19.05
1.00	25.4
1.03	26.2
1.19	30.2
1.31	33.3
3.375	85 72
3.781	96.04
5.88	149.4

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3 Solid line indicates location, shape, and dimensions for minimum rejection groove
4. Dashed line indicates location, shape, and dimensions for maximum rejection groove.

FIGURE 1. Classes RK1 and RK5 fuses, 250 volts, 61-100 amperes.

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.
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REQUIREMENTS

Design and construction See figure 1

Physical Nonrenewable.

Terminals Knife blade type.

Material. Copper or copper alloy.

Strength 5 inch-pound torque between body and blades.

Body

Insulating material Fiber, ceramic, melamine-impregnated glass fiber, or other suitable material

Electrical

Voltage 250 volts or less.

Frequency 48 to 60 hertz.

Current See table I.

Current carrying capacity 110 percent indefinitely with temperature rise not to exceed 30°C (90°F) above ambient on the body and the blades.

Overload interrupt

Instantaneous. Shall interrupt within 2 hours at 135 percent of rated current and within 6 minutes at 200 percent of rated current.

Time delay In addition to the above requirement, the fuse shall not interrupt 500 percent of rated current within 10 seconds.

Interrupting capacity rating 200,000 amperes rms symmetrical at 250 volts, 48 to 60 hertz and a power factor of 20 percent or less. Closing angle shall be essentially at the zero of the voltage wave (maximum offset) or later, to produce start of arcing within 30 electrical degrees prior to system peak voltage

Threshold ratio 30 maximum for RK1, 65 maximum for RK5.

Peak let-thru current See table II

Maximum clearing I^2T See table II.

Applicable fusenolder Part number W-F-870/8-01 of W-F-870/8.

Government part number The Government part number shall consist of the prefix "WF1814/58-" followed by the part number designation shown in table I.

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TABLE I. Current rating and part number designation

Current in amperes	Part number designation			
	RK5		RK1	
	Instantaneous	Time delay	Instantaneous	Time delay
70	F5070	D5070	F1070	D1070
80	F5080	D5080	F1080	D1080
90	F5090	D5090	F1090	D1090
100	F5100	D5100	F1100	D1100

TABLE II. Peak let-thru and I^2T current.

	Between threshold and 50 kA		100 kA		200 kA	
	RK1	RK5	RK1	RK5	RK1	RK5
Maximum acceptable peak let-thru current ($I_p \times 10^3$)	14	22	15	25	20	32
Maximum clearing I^2T (amperes squared seconds) ($I^2T \times 10^3$)	100	500	100	500	100	500

MILITARY INTERESTS

Custodians

Army - ER
Navy - YD
Air Force - 85

Review Activities

Army - AL, AT, AZ
Air Force - 99
DLA - ES
NS

User Activities

Army - AR
Navy - OS

CIVIL AGENCY COORDINATING ACTIVITIES

DOE - DOE
COMMERCE - NBS
DOT - ACO, APM
GPO - GPO
GSA - FSS
JUSTICE - FBI
NASA - JFK, LRC
USPS - POS
VA - OSS

PREPARING ACTIVITY

DLA - ES

(Project 5920-0434-03)

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